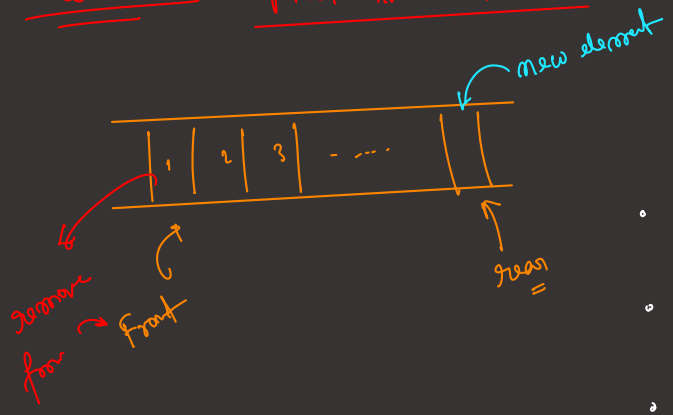
Queue — First in First Out

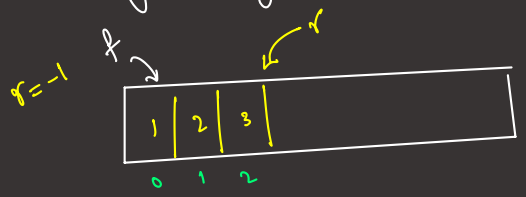


insertion order = deletion order

- insert / add / enqueue O(1)
- deletion / remove / dequeue O(1) Deque DS
- Peak() → Front O(1)

Implementation of Queue

(i) using Array

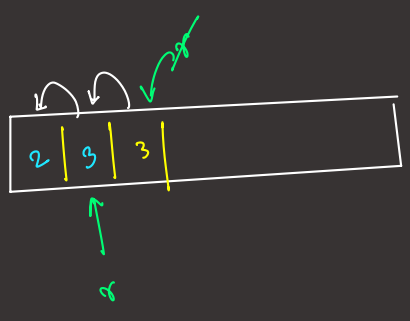


O(1)

insert

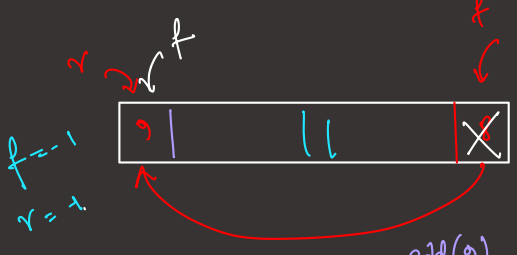
[1, 2, 3]

remove



O(n)

O(1) → Circular Queue



add(9)

add(1)
add(2)
add(3)
remove() → 1

rear = (rear+1) % size

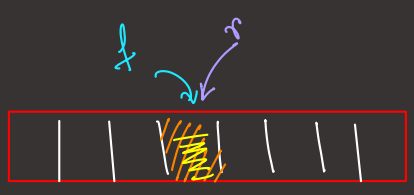
front = (front+1) % size

full → { r - f = size-1 }

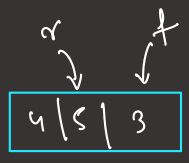
(r+1) % size = f

condition to apply

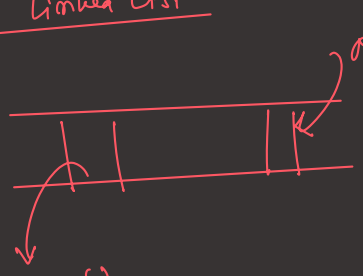
{ f=-1, r=-1 }

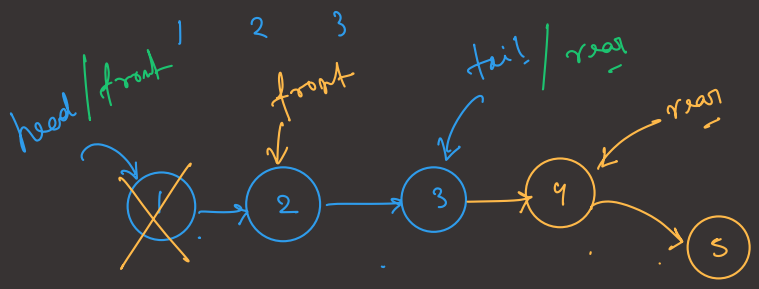


remove



Queue using Linked List





Queue Using JCF

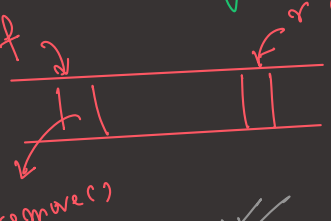
Queue ← interface.

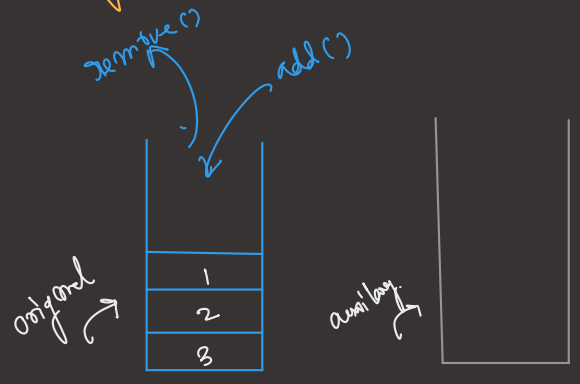
```
Queue<Integer> q1 = new LinkedList<>();
Queue<Integer> q2 = new ArrayDeque<>();
```

[1 2 3]

Question

implement a Queue using two Stack





H.W

way 1

add → O(1)

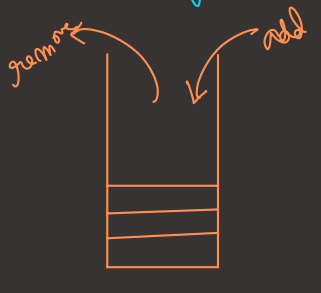
and remove → O(n)

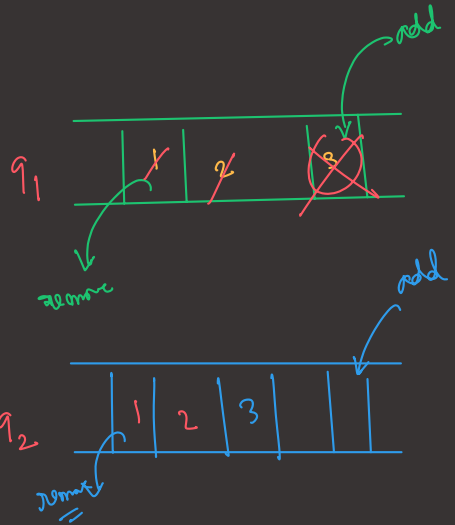
or way 2

add → O(n)

remove → O(1)

Stack Using two Queue





H.W

way 1

push → O(1)

pop → O(n)

way 2

push → O(n)

pop → O(1)

```
pop() {  top = -1
if (!q1.isEmpty()) {
  while (!q1.isEmpty()) {
    top = q1.remove();
    if (q1.isEmpty()) {
      break;
    }
    q2.add(top);
  }
} else {
  while (!q2.isEmpty()) {
    top = q2.remove();
    if (q2.isEmpty()) {
      break;
    }
    q1.add(top);
  }
}
return top
```

TC = O(n)

↑

pop, peek

Question

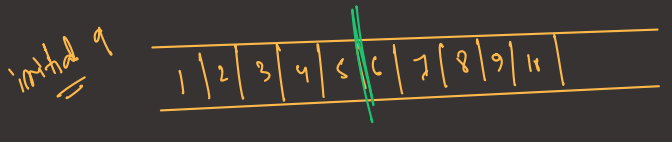
interleave 2 halves of a Queue. (even length)

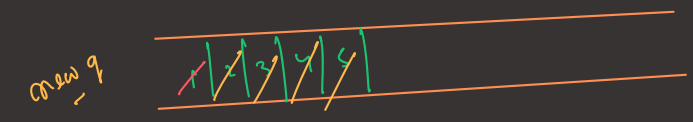
given: 1 2 3 4 5 | 6 7 8 9 10

o/p: 1 6 2 7 3 8 4 9 5 10

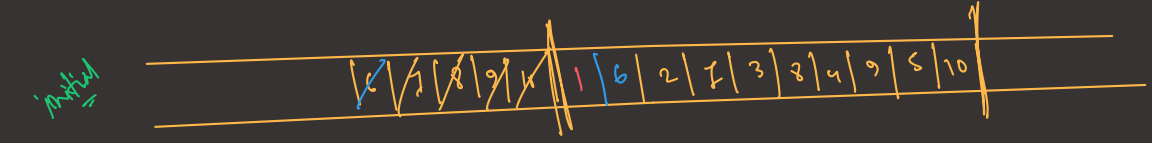
upto q.size()/2 from initial queue

remove and put it into the new queue






```
while (!newq.isEmpty()) {
  initialq.add(newq.remove());
  initialq.add(initialq.remove());
}
```



Question

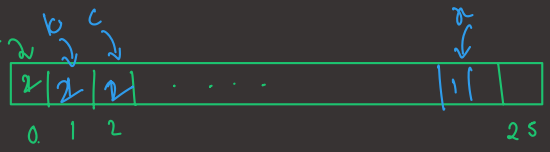
First Non-repeating Letter in a Stream of character

↳ all character will be in small letter

Stream = a a b c c x b

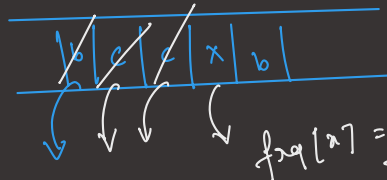
	o/p
a	a
a a	-1
a a b	b
a a b c	b
a a b c c	b
a a b c c x	b
a a b c c x b	x

freq map[26] =



idx → 'ch' - 'a'

freq[a] = 2



freq[x] = 1

peek

freq. != 2

o/p = a

= -1

= b

= b

= b

= b

= x

queue<char> q

for (int i = 0 to str.length) {

ch = str.charAt(i)

q.add(ch) , map update

q.peek()

front

map[front] → frequency > 1

q.remove()

ch [freq] = 1 → ch → first non-repeating letter

q → empty() return = -1

① freq[] = [26] 'a' → 'z'

② Queue<Character> q

③ for(i=0 to str.length) {

ch = str.charAt(i);

q.add(ch);

freq['ch' - 'a']++;

q.remove() → till I found 1st non repeating letter

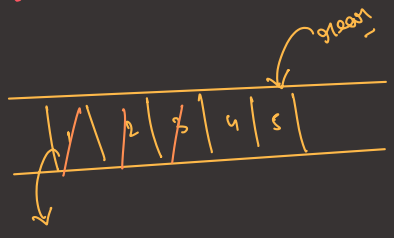
}

Question

Queue Reversal

Queue → [1 2 3 4 5] ; print in reverse order

o/p : 5 4 3 2 1



operation

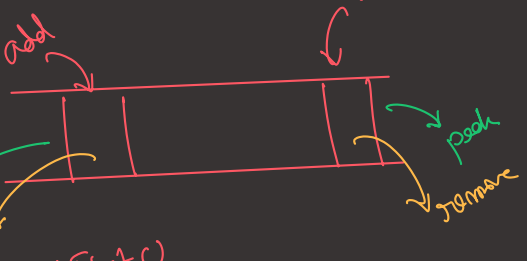
remove() → O(1)

5 4 3 2 1

Dequeue

DS → Double ended queue

Deque



addFirst()

addLast()

removeFirst()

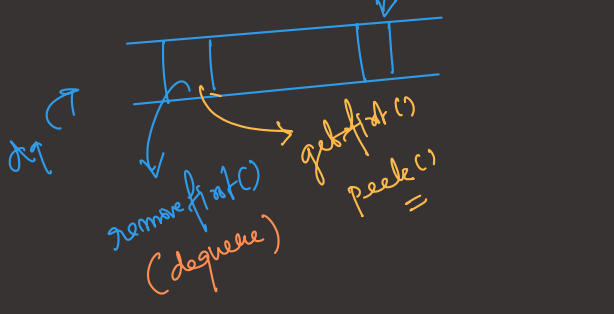
removeLast()

getFirst()

getLast()

Deque<Integer> dq = new LinkedList<>();

Queue using Deque



Stack using Deque